

STM32CubeIDE Setup Guide for MRSUBG Tx/Rx Wake-Up Radio Project

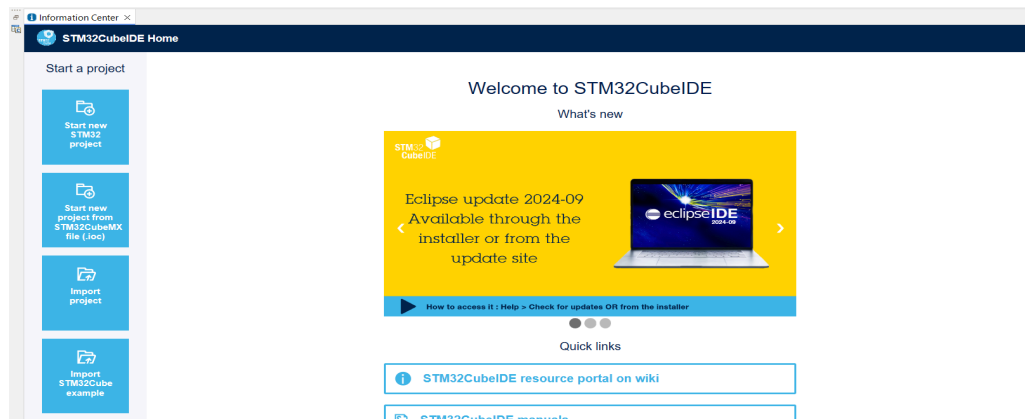
 STM32CubeIDE Version: 1.18.0

 Project: Wake-Up Radio (LPAWUR) with MRSUBG_BasicGeneric_Tx/Rx Example

Step-by-Step Instructions

1. Launch STM32CubeIDE

- After opening the software, **select or create a new workspace**.



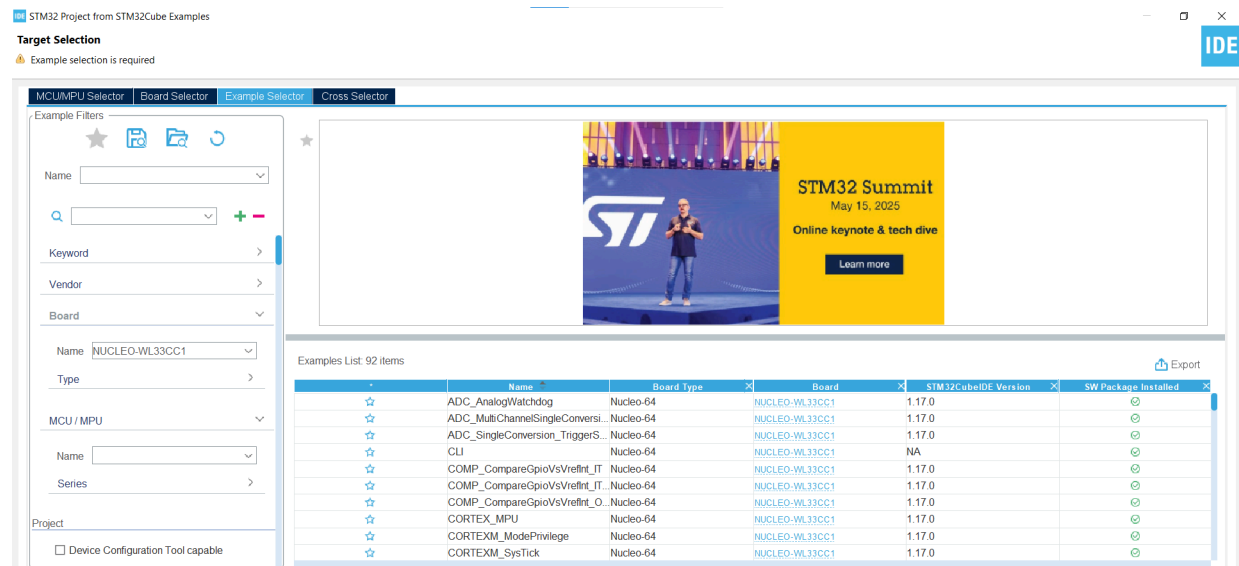
2. Import the STM32Cube Example

- At the bottom-left corner, click on **“Import STM32Cube Example”**.



3. Select Your Board

- In the search bar under “**Board**”, enter your board name (e.g., **NUCLEO-WL33CC1**).



4. Choose Example Project

- A list of available examples will appear.

Examples List: 92 items

Name	Board Type	Board	STM32CubeIDE Version	SW Package Installed
LPAWUR_Skeleton	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓
LPAWUR_WakeupRadio_Rx	Nucleo-64	NUCLEO-WL33CC1	NA	✓
LPUART_TwoBoards_CoMT	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓
LPUART_WakeUpFromDeepStop...	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓
MRSUBG_802_15_4_Rx	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓
MRSUBG_802_15_4_Tx	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓
MRSUBG_BasicGeneric_Rx	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓
MRSUBG_BasicGeneric_Tx	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓
MRSUBG_Chat	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓
MRSUBG_DatabufferHandler_Rx	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓

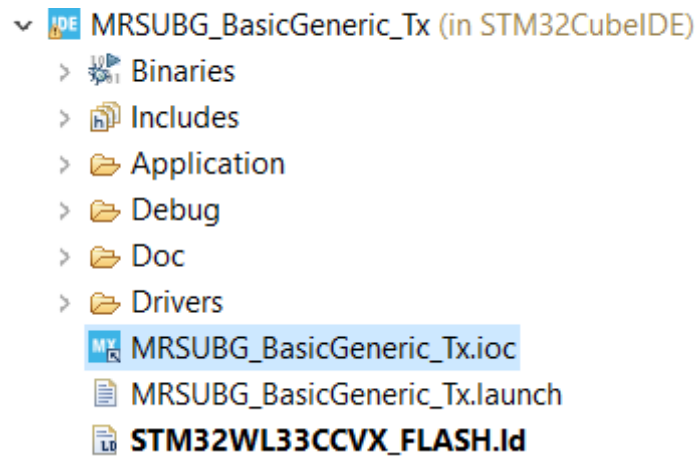
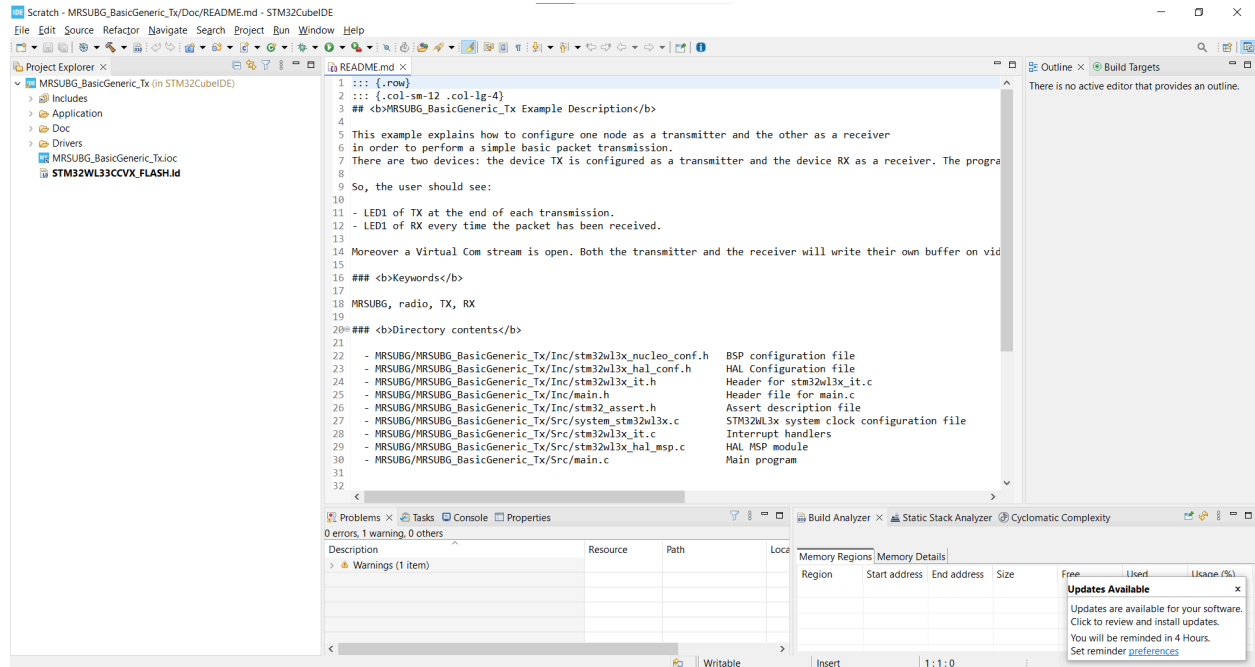
- Search for: **MRSUBG_BasicGeneric_Tx** or **MRSUBG_BasicGeneric_Rx**

MRSUBG_BasicGeneric_Rx	Nucleo-64	NUCLEO-WL33CC1	1.17.0
MRSUBG_BasicGeneric_Tx	Nucleo-64	NUCLEO-WL33CC1	1.17.0

- Click on your chosen project and then click **Next > Finish**.

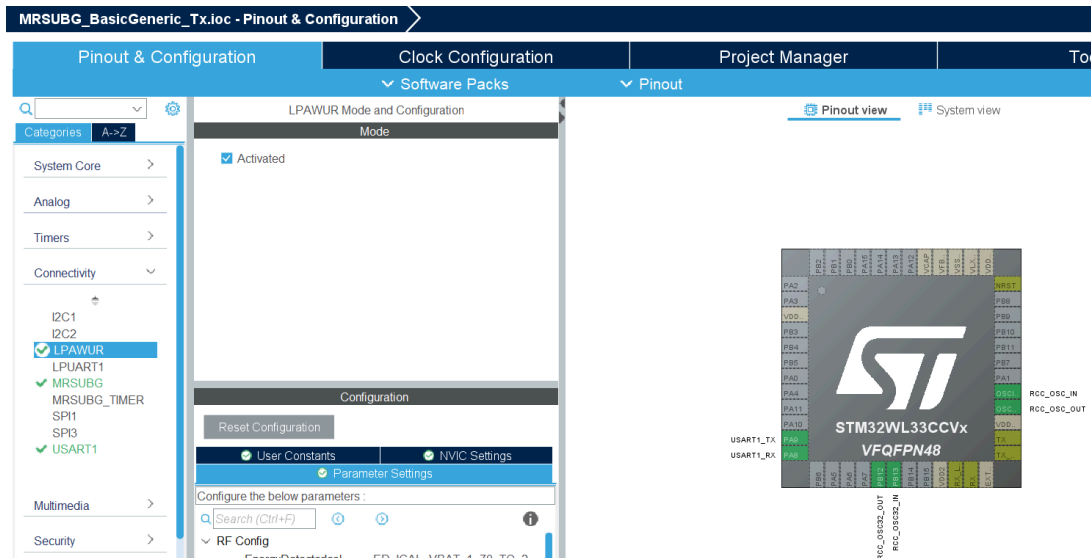
5. Access the Project IOC Configuration

- Once the project is open, locate and double-click the **.ioc** file.



6. Enable Wake-Up Radio Feature

- In the IOC editor:
 - Go to **Connectivity**, and activate **LPAWUR** (Low Power Wake-Up Radio).
 - Save with **Ctrl + S**.



7. Add Your Source Code

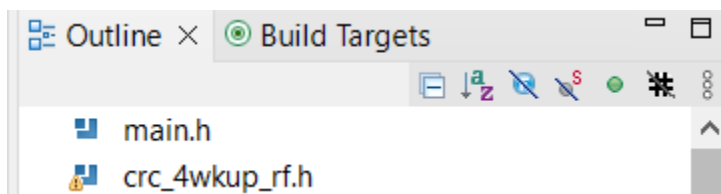
- Go to [Application/User/main.c](#) and paste the code.

```

main.c  README.md  main.c  main.h  main.h  Scratch_Tx_Rx.ioc  main.c
44  /* USER CODE BEGIN PV */
45  __IO uint32_t aRandom16bit[100];
46  __IO int32_t rssi = 0;
47  uint8_t LPAWUR_Payload[8];
48  /* USER CODE END PV */
49  /* Private function prototypes -----*/
50  void SystemClock_Config(void);
51  void PeriphCommonClock_Config(void);
52  static void MX_GPIO_Init(void);
53  static void MX_MRSUBG_Init(void);
54  /* USER CODE BEGIN PFP */
55  static void RandomNumbersGeneration(uint8_t j);
56  void CreateLPAWURFrameV2(uint8_t j,uint8_t aRandom16bit);
57  void EvaluateCrc(uint8_t * LPAWUR_payload);
58  void UTIL_LPM_EnterLowPower(void);
59  void UTIL_LPM_Init(void);
60  void MX_APPE_Process(void);
61  void GotoRx(void);
62  void MX_APPE_Idle(void);
63  static void MX_LPAWUR_Init(void);
64  void UTIL_LPM_Init( void );
65  /* USER CODE END PFP */
66  /* Private user code -----*/
67  /* USER CODE BEGIN 0 */
68  /* USER CODE END 0 */
69  /**
70   * @brief The application entry point.
71   * @retval int
72   */
73  int main(void)
74  {
75  /* USER CODE BEGIN 1 */

```

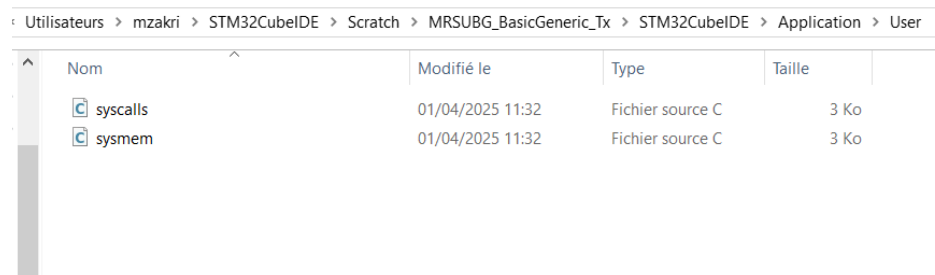
- To find [main.h](#) double click on the file on the top -left corner and paste the code.



8. Copy Additional Folders (LPM and WUR)

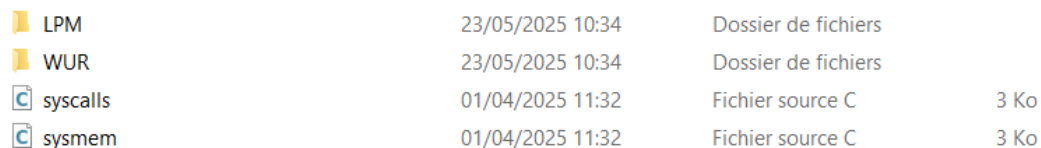
- These folders contain essential drivers for the wake-up functionality.
- Navigate to your workspace folder (e.g.,

`C:\Users\<YourName>\STM32CubeIDE\Scratch\MRSUBG_BasicGeneric_Tx\STM32CubeIDE\Application\User`).



Nom	Modifié le	Type	Taille
syscalls	01/04/2025 11:32	Fichier source C	3 Ko
sysmem	01/04/2025 11:32	Fichier source C	3 Ko

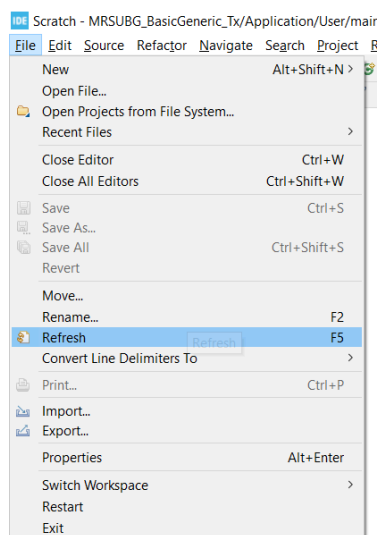
- Copy the **LPM** and **WUR** folders into your project directory under the **User** folder



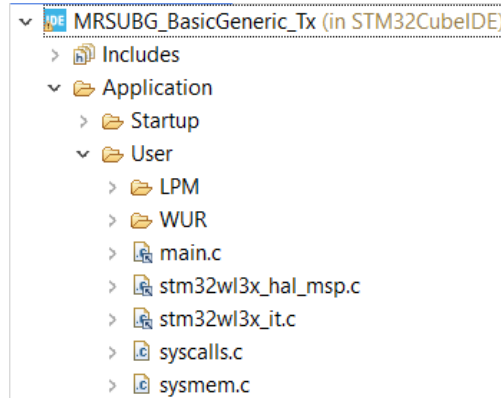
LPM	23/05/2025 10:34	Dossier de fichiers	
WUR	23/05/2025 10:34	Dossier de fichiers	
syscalls	01/04/2025 11:32	Fichier source C	3 Ko
sysmem	01/04/2025 11:32	Fichier source C	3 Ko

9. Refresh the IDE

- After copying, close the file explorer and return to STM32CubeIDE.
- **File > Refresh.**



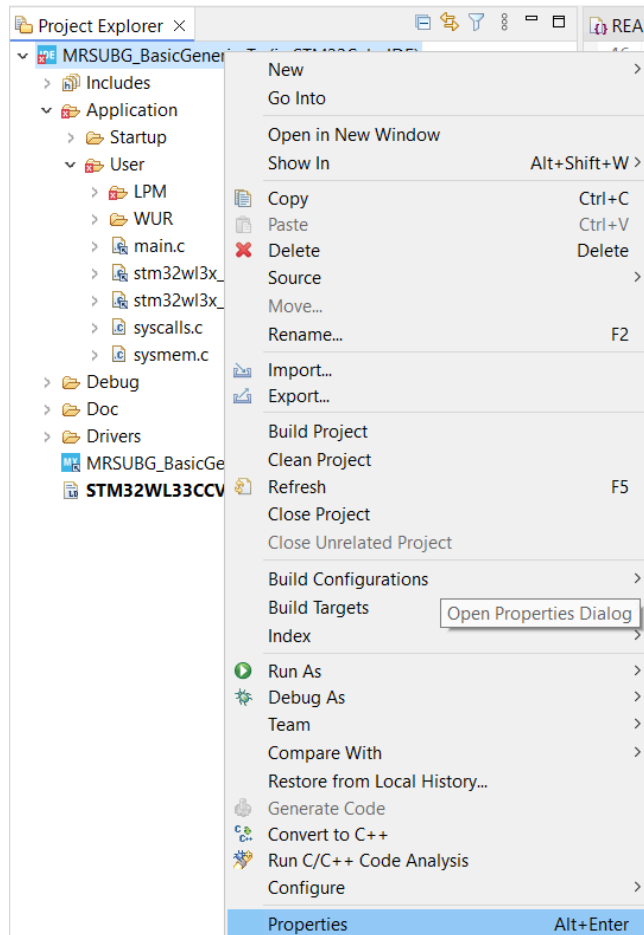
- The folders should now appear in the Project Explorer.



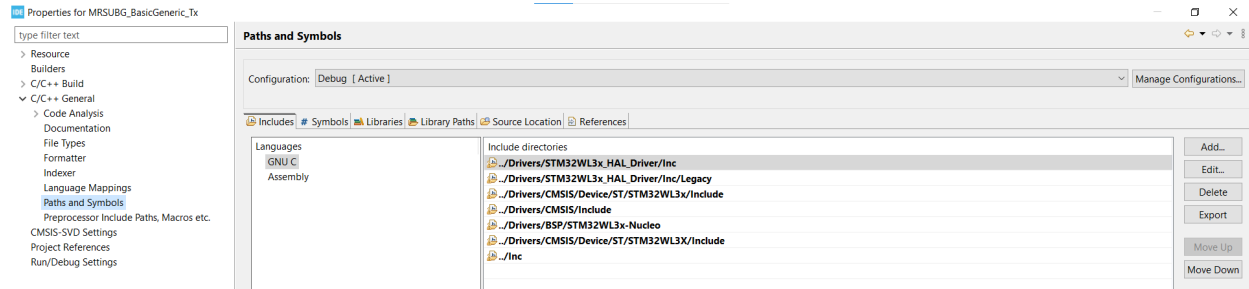
10. Add Include Paths

🔧 Without this step, the compiler won't recognize the new folders.

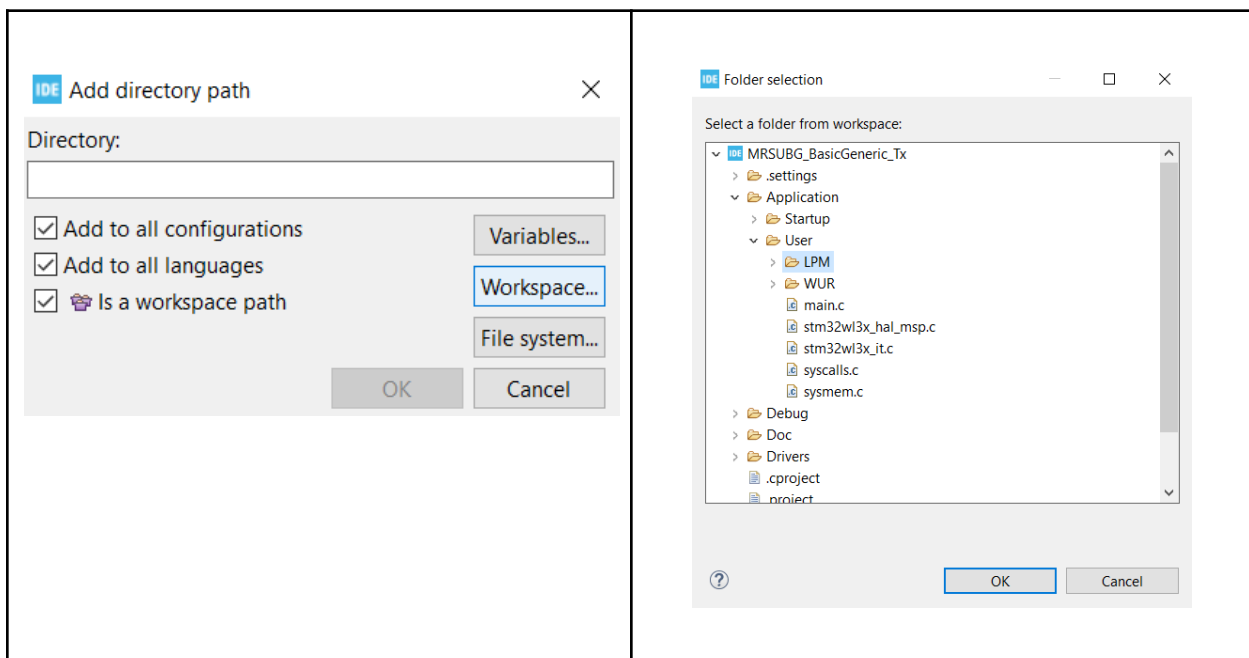
- **Right-click the project > Properties.**



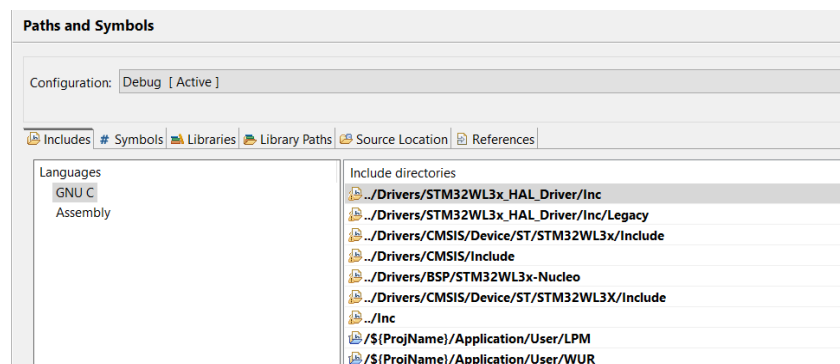
- Go to **C/C++ General > Paths and Symbols**.



- Under the **Includes** tab, click **Add...** > **Workspace** > select **LPM**, check all boxes > OK.



- Repeat for the **WUR** folder.



- Apply and close the settings, then **refresh the IDE again**.

11. Build the Project

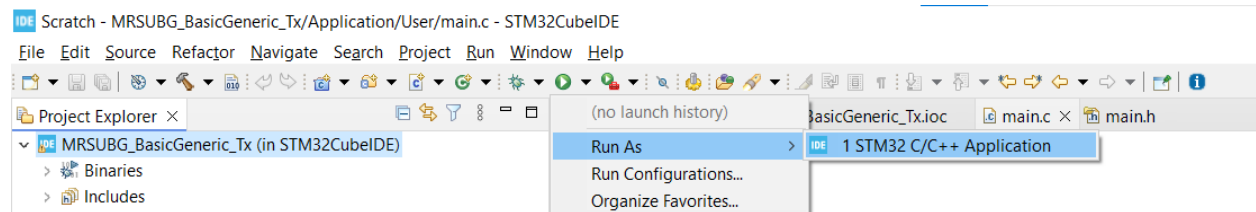
- Click on the **hammer icon**  to build.
- Warnings may appear the first time but **should disappear on the second build**.

```
text      data      bss      dec      hex filename
31336     156     5736    37228    916c MRSUBG_BasicGeneric_Tx.elf
Finished building: default.size.stdout
```

10:48:05 Build Finished. 0 errors, 0 warnings. (took 330ms)

12. Run the Program

- Click the **green triangle icon** to start debugging.
- Then click **STM32 C/C++ Application** then **OK** to run.



13. Select Tx or Rx Mode

- Open `main.c` and change the value of the `mode` variable to:
 - `MODE_TX` for Transmit : 0
 - `MODE_RX` for Receive : 1

```
12 uint8_t mode = 0; // Rx = 1 , Tx = 0
```

 Make sure both transmitter and receiver boards are powered and flashed accordingly.

Troubleshooting Tips

Issue	Solution
Board not showing up	Ensure your USB is connected, try restarting STM32CubeIDE
Project fails to build	Make sure LPM/WUR paths are included, and files are properly copied
Warnings during build	Ignore if they appear on first build — recompile
No output observed	Confirm Tx and Rx boards are flashed with corresponding roles

You're Done!

You now have a working setup for low-power wake-up radio communication using STM32CubeIDE and the STM32WL Nucleo boards.

Below is an example of a Tx to Rx :

```
COM3 - PuTTY
STM32WL3 LPAWUR - Transmitter example.
0 5
Number of packet received 0
█
```